

Problem 1:

a) You are given a scalar field f and a vector field $\vec{A}$, such that $\vec{A} = \nabla f$. What is $\nabla \times \vec{A}$?

$$\nabla \times \vec{A} = 0 \quad \vec{A} \text{ is conservative, as } \nabla \times (\nabla f) = 0$$

For each of the two vector fields in b) and c) below, calculate the curl and divergence. Then circle the correct answer to whether the field is conservative, solenoidal, both, or neither.

b) $\vec{A} = \hat{x}(2x^2 + 3z^2) - \hat{y}(2y^2 + 3x^2) + \hat{z}(2x^2 + 5y^2)$

conservative

solenoidal

both

neither

$$\nabla \cdot \vec{A} = 4x - 4y + 4z \neq 0$$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ 2x^2 + 3z^2 & -(2y^2 + 3x^2) & 2x^2 + 5y^2 \end{vmatrix} =$$

$$= \hat{x} 10y - \hat{y} (-10x) + \hat{z} (-10x)$$

$$= \hat{x} 10y + \hat{y} 10x - \hat{z} 10x \neq 0$$

c) $\vec{A} = \hat{R} 5 \exp(-10R)/R$

conservative

solenoidal

both

neither

$$\nabla \times \vec{A} = 0$$

Only A_R component, & depends only on R .

$$\begin{aligned} \nabla \cdot \vec{A} &= \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 5 \frac{e^{-10R}}{R} \right) \\ &= \frac{1}{R^2} (5e^{-10R} - 50R e^{-10R}) = \frac{5e^{-10R}}{R^2} (1 - 10R) \end{aligned}$$

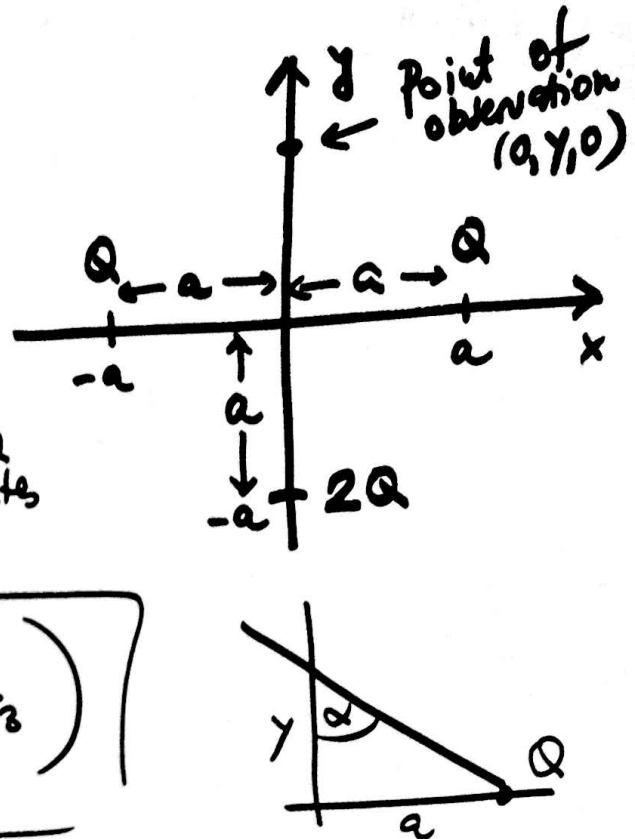
Problem 2:

a) Calculate the electric field $\vec{E}$ along the y axis due to the distribution of point charges depicted on the right.

Assume the surrounding medium is air (permittivity ϵ_0).

x-components due to the two Q 's cancel

$$\begin{aligned}\vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{Q}{(y^2+a^2)^{3/2}} \frac{y}{\sqrt{y^2+a^2}} \hat{y} \cdot 2 \\ &\quad + \frac{1}{4\pi\epsilon_0} \frac{-2Q}{|y+a|^3} (\cos\alpha) \hat{y} \quad \text{each } Q \text{ contributes} \\ \vec{E} &= \hat{y} \frac{Q}{2\pi\epsilon_0} \left(\frac{y}{(y^2+a^2)^{3/2}} - \frac{y+a}{|y+a|^3} \right)\end{aligned}$$



b) Calculate the electric potential V along the y axis. Assume $V(0)=0$.

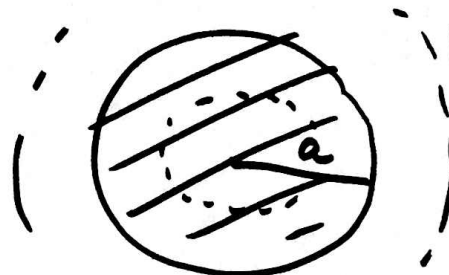
Using the principle of superposition

$$\begin{aligned}V &= \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{y^2+a^2}} \times 2 \\ &\quad + \frac{1}{4\pi\epsilon_0} \frac{-2Q}{|y+a|} = \frac{Q}{2\pi\epsilon_0} \left(\frac{1}{\sqrt{y^2+a^2}} - \frac{1}{|y+a|} \right)\end{aligned}$$

Problem 3: A sphere of radius a has uniform volume charge density ρ_{v0} . There is no charge outside of the sphere. Assume the permittivity of the sphere and the surrounding medium is ϵ_0 .

a) Calculate E everywhere in space.

Gaussian surfaces are spheres.
 $4\pi R^2 D(R) = \int \rho_v dV = \rho_{v0} \frac{4}{3} R^3 \pi$
↑
inside of sphere
of radius $R < a$



$R < a$:

$$D(R) = \frac{\rho_{v0} R}{3}, R < a$$

$R > a$: $4\pi R^2 D(R) = \rho_{v0} \frac{4}{3} a^3 \pi$
all of the sphere charge

$$D(R) = \frac{\rho_{v0} a^3}{3R^2}, R > a$$

$$R < a, \vec{E} = \hat{R} \frac{\rho_{v0} R}{3\epsilon_0}$$

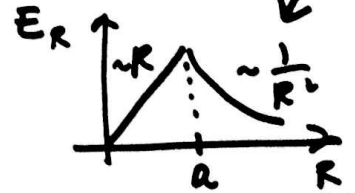
$$R > a, \vec{E} = \hat{R} \frac{\rho_{v0} a^3}{3\epsilon_0 R^2}$$

b) Calculate the electric potential V , assuming it is zero at an infinitely large distance from the sphere.

$$V(\infty) = 0$$

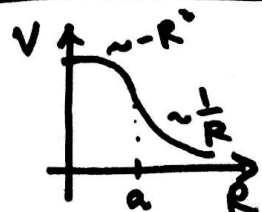
$$R > a \quad V(R) - V(\infty) = - \int_{\infty}^R \vec{E} \cdot d\vec{l} \quad (d\vec{l} = \hat{R} dl)$$

$$= - \int_{\infty}^R E(R) \cdot dR$$



$$= - \frac{\rho_{v0} a^3}{3\epsilon_0} \left(-\frac{1}{R} \right) \Big|_{\infty}^R = \frac{\rho_{v0} a^3}{3\epsilon_0 R} = V(R > a)$$

$$R < a \quad V(R) - V(a) = - \int_a^R \frac{\rho_{v0} R}{3\epsilon_0} dR = - \frac{\rho_{v0}}{3\epsilon_0} \left(\frac{R^2 - a^2}{2} \right)$$



know this from the $R > a$ integral
 $V(a) = \rho_{v0} a^2 / 3\epsilon_0$

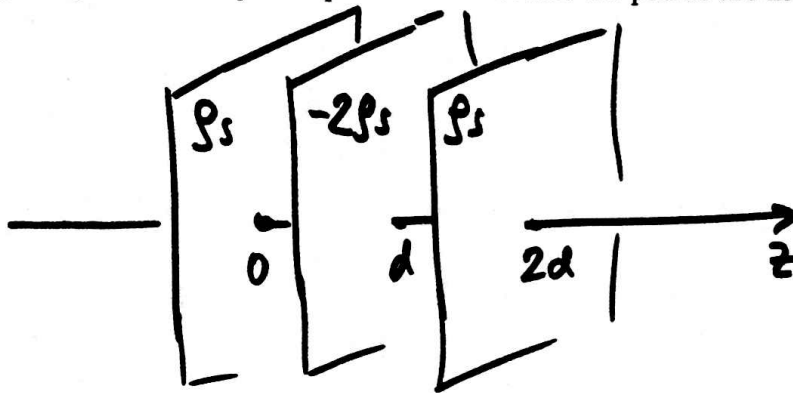
$$V(R) = \frac{\rho_{v0} a^2}{3\epsilon_0} + \frac{\rho_{v0} a^2}{6\epsilon_0} - \frac{\rho_{v0} R^2}{6\epsilon_0}$$

$$\left(\frac{1}{3} + \frac{1}{6} = \frac{1}{2} \right)$$

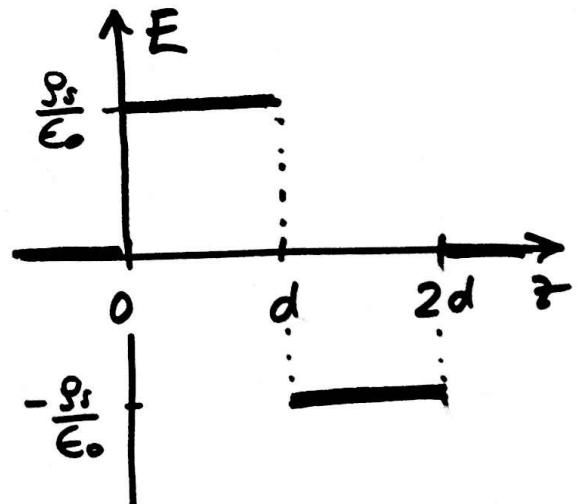
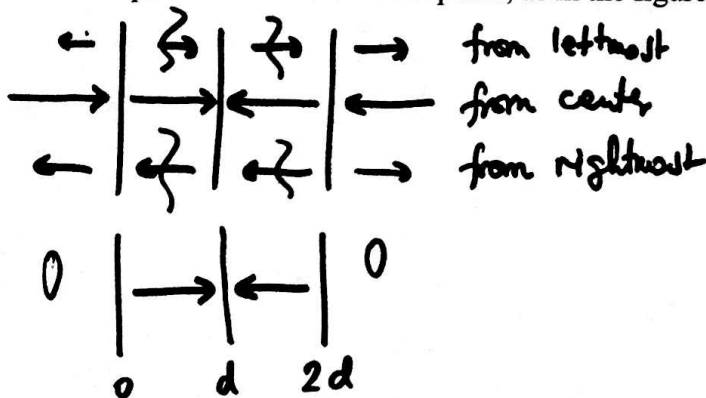
$$V(R < a) = \frac{\rho_{v0} a^2}{2\epsilon_0} - \frac{\rho_{v0} R^2}{6\epsilon_0}$$

Problem 4:

Three parallel infinite planes have surface charge densities ρ_s , $-2\rho_s$, and ρ_s ($\rho_s > 0$). The spacing between adjacent planes is d . Assume the planes are in vacuum (permittivity ϵ_0).

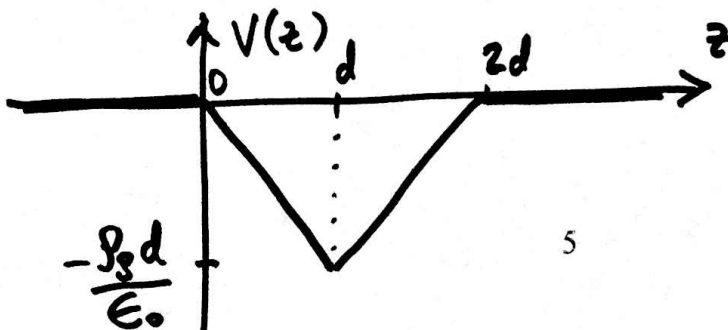


a) Calculate and graph the electric field E everywhere in space. Place the coordinate origin at the position of the leftmost plane, as in the figure above.



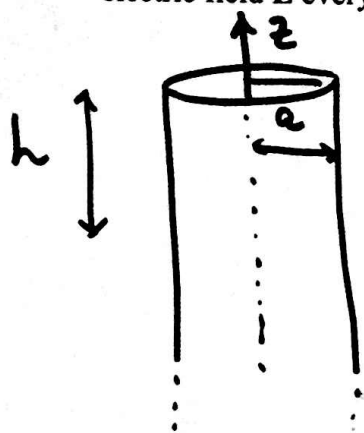
b) Calculate and graph the electric potential V everywhere in space. As the reference point, use the leftmost plane, i.e., assume $V(0)=0$.

The integral from $-\infty$ to $\text{any } z < 0$ is 0, so $\underline{V(z < 0) = 0}$
 The integral from 0 to $2d$ is 0, as $+$ & $-$ regions have the same width, so $V(2d) = 0$.
 Furthermore $V(z > 2d) = V(2d)$ (field zero), so $\underline{V(z > 2d) = 0}$



Problem 5:

a) A cylindrical shell of line charge density ρ_l has radius a . Use Gauss's law to calculate the electric field $\mathbf{E}$ everywhere in space. Assume the surrounding medium is air (permittivity ϵ_0).



$r < a$ (inside shell) $\mathbf{D}(r) = 0, \mathbf{E}(r) = 0$

$r > a$ $2\pi r h \mathbf{D}(r) = Q \leftarrow \text{all the } Q \text{ on cylinder of height } h$

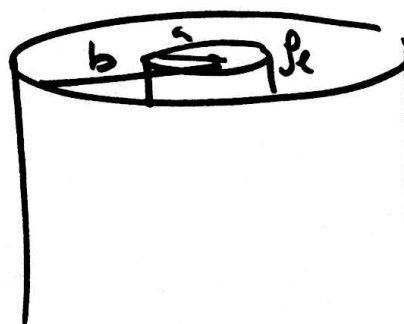
$\mathbf{D}(r) = \frac{1}{2\pi r} \left(\frac{Q}{h} \right) = \rho_l$

$r < a : \mathbf{E} = 0$

$r > a : \mathbf{E} = \hat{r} \frac{\rho_l}{2\pi\epsilon_0 r}$



b) Another cylindrical shell, with the line charge density of $-\rho_l$ and radius $b > a$ is placed coaxially around the shell from part a). Based on your result from a) and using the principle of superposition, calculate and graph the total electric field everywhere in space.

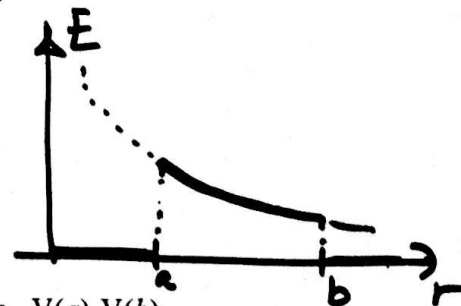


$r < a : \mathbf{D} = 0$
 $r > b : \mathbf{D} = 0$ (total charge enclosed is zero)

$a < r < b$: enclosed only from inner cylinder.

$\mathbf{D}(r) = \rho_l / 2\pi r$

$\mathbf{E}(r < a) = \mathbf{E}(r > b) = 0$
 $\mathbf{E}(a < r < b) = \hat{r} \frac{\rho_l}{2\pi\epsilon_0 r}$



c) Calculate the potential difference between the inner and outer shells, i.e., $V(a) - V(b)$.

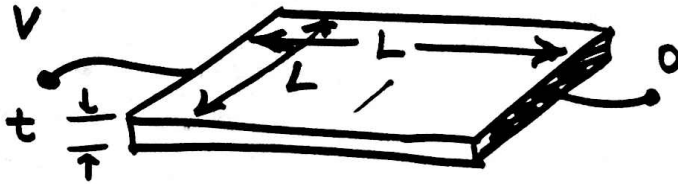
$V(a) - V(b) = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = \int_a^b E(r) dr = \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$

$V(a) - V(b) = \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$

Problem 6:

A copper sheet of thickness t has square faces (5 cm by 5 cm).

- a) Assume bias is applied between the opposite edges of a square face. Calculate the resistance R . (Conductivity of copper can be found in the table).

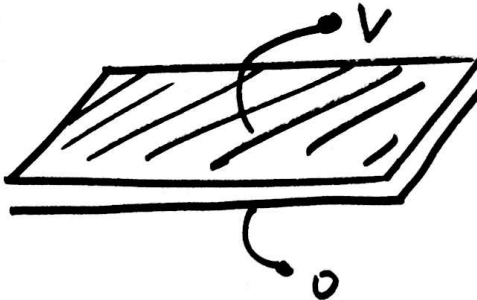


$$R = \frac{1}{\sigma} \frac{l}{A} \quad \begin{matrix} l = L \\ A = Lt \end{matrix}$$

$$R = \frac{1}{\sigma t} = \underline{\underline{1.72 \times 10^{-5} \Omega}}$$

$$\sigma = 5.8 \times 10^7 \text{ S/m}$$

- a) Assume bias is applied between the two square faces. Calculate the corresponding resistance R .



$$R = \frac{1}{\sigma} \frac{l}{A} \quad \begin{matrix} A = L^2 \\ l = t \end{matrix}$$

$$R = \frac{1}{\sigma} \frac{t}{L^2}$$

$$R = 6.9 \times 10^{-9} \Omega$$